Research Dossier — Phase 1 (English)

Comprehension and feasibility of S, F, and Q — English

Operational summary · v2.2 · November 8, 2026
Status: public draft; not submitted or preregistered.

In one line: proposed budget of **US$5,000**, an **11-month** planning structure, and up to **38 participants** (up to 18 in cognitive interviews and 20 in the separate pilot).

According to the reference documents, there has been no ethics approval, recruitment, data collection, human result, submission, or publication. No participant activity will begin before the applicable ethical and financial conditions are met.

This Phase 1 **does not test the F×Q–S association, does not validate F, Q, or S**, and does not provide a basis for the sample size of a confirmatory study.

Section 1 — Question, concepts, and limits

- **F**: mental content oriented toward future plans, possibilities, or events.
- **Q**: sequential or chain-like structure, in which one idea leads to the next.
- **S**: a report of thought content or functions that the person recognizes as qualitatively distinct and perceives as present at the same time. Examples of qualities may include planning steps, an emotion, a memory, or evaluating the meaning of a situation. Words and images are modalities—forms of presentation—and do not count as distinct dimensions on their own.

The long-term hypothesis is **conditional and observational**: at high Q, higher F could be associated with a lower probability of provisionally reporting S≥2; at low Q, this slope would be smaller or close to zero. The proposed moderation is motivated by the hypothesis that maintaining a future-oriented chain may require keeping intermediate steps and their relations in mind, unlike an isolated future-oriented thought. Working Memory, Representational Space, links to neural architecture, and possible age effects are candidate mechanisms or hypotheses, not demonstrated results. The design does not support inferences of causality or parallel neural processing.

The proposed recall period covers the previous two hours. Responses may mix or reconstruct episodes; this is not a direct measure of a theoretical seconds-scale dynamic.

Phase 1 **does not test F×Q–S or validate the measures**. Interviews and vignettes check comprehension of the proposed definition; they are not sufficient to validate S or demonstrate discriminant validity.

Section 2 — Design and participants

The two components are separate and do not share participants.

Component	Cap and design	Purpose
Cognitive interviews	Up to 18 people aged 18–65, in stages of 6 + 6 + up to 6	Compare two S wordings and assess comprehension of S, F, and Q.
Pilot	20 adults aged 18–65 in Brazil, recruited by convenience; three consecutive days	Describe comprehension, completion, burden, missing data, ordinal distributions, and within-person variation.

The pilot will have five scheduled responses per day: **15 windows per person, 300 windows in total, and up to 900 ordinal responses** (F, Q, and S in each window). The combined cap is up to 38 participants. These are feasibility-planning numbers, not a power calculation.

Section 3 — Interviews, items, and vignettes

In the first round, six people assess S-A or S-B (three per wording), as well as the F and Q items. The choice of S wording will be qualitative: favor the wording with fewer recorded confusions; in a tie, choose S-A. Another six people assess the selected wording. If needed, one revision and retest with up to six additional people is allowed within the same cap of 18. The final wording is frozen before the pilot.

Provisional items and anchors

The initial response referent will be **frequency of occurrence within the interval**. All items below provisionally use the same format; the wording and anchors are candidate instrument choices to be tested in the interviews before the instrument is frozen. They are not a validated scale.

- **S-A:** “During the previous two hours, how often did you notice thought content or functions that were qualitatively distinct and present at the same time?”
- **S-B:** “During the previous two hours, how often did your experience simultaneously include more than one quality of thought that differed in nature?”
- **F:** “During the previous two hours, how often was your thinking oriented toward future plans, possibilities, or events?”
- **Q:** “During the previous two hours, how often did your ideas form a sequence, with one leading to the next?”

Candidate anchors: **Never / Rarely / Sometimes / Frequently / Almost always**. The person will first be invited to explain S in their own words. The rubric requires two aspects of different nature to be present at the same time. S is a subjective self-report rule, not an objective taxonomy; modality (words/image) alone is not enough, and any example, including an emotion, counts only if the person reports those two aspects as simultaneous. Two raters code the responses and resolve disagreements by consensus.

Vignettes and provisional key

1. While rehearsing what to say in a difficult conversation, the person notices, at the same moment, an emotional concern about how the other person will respond. → Co-presence only if the person reports planning and concern as two aspects that differ in nature and are present at the same time.
2. While assembling a piece of furniture, the person follows the current step and, at the same time, remembers a past assembly that prompts a comparison. → Co-presence only if the person reports the task and the memory as two aspects that differ in nature and are present at the same time.
3. The person first thinks about a future meeting and then turns to shopping; each topic replaces the previous one. → Not co-presence: successive content.
4. While planning a trip, one idea leads to the next—time, transport, route—with no other dimension present alongside it. → Not co-presence: Q can be high without S.
5. The person describes a single thought presented in words and images. → Not co-presence: different modalities described as one thought.

The individual criterion is at least 4 correct out of 5, including one co-presence vignette and one non-co-presence vignette, in addition to the spontaneous explanation. The key is set before the interviews; rater disagreements are resolved by consensus.

Section 4 — Pilot and administration

Each item asks about the immediately preceding two hours and is answered at the end of the window. The frequency operationalization measures recurrence within the interval, not duration, intensity, or proportion of time (nor cognitive load); any future F×Q–S hypothesis must retain this operationalization or redefine it before testing. The proposed local-time windows, the same on all three days, are **08:00–**

10:00, 10:30–12:30, 13:00–15:00, 15:30–17:30, and 18:00–20:00. The windows do not overlap; final times still require confirmation.

F, Q, and S are answered separately. The provisional administration format is a **scheduled diary form by window**; the analysis will be descriptive, with the **analysis language to be determined before data collection**. The instrument, times, anchors, and final specification must be set before data collection; no specific platform is assumed.

Among participants who consent, six will be selected by lottery before data collection for four brief probes, “Right now, what is on your mind?”, on the second day. These are outside the 300 windows and will be summarized separately. Because they occur before the retrospective response, they may anchor recall; they are not validation and are not temporally equivalent to the two-hour recall period.

Section 5 — Descriptive plan

The scale is ordinal; equal distances between categories are not assumed. The full distributions of S, F, and Q will be reported, with counts and proportions in every category.

S≥2 will provisionally mean the second category or above, named **“Rarely or more”** according to the candidate anchor. It is only a descriptive summary. For grouped descriptions, high F will be Frequently/Almost always and low Q will be Never/Rarely. For each combination of F/Q categories, always show the numerator and denominator for S≥2; calculate the cell proportion when n>0; mark n=0 as unavailable.

Completion will be described overall and by participant. For within-person variation, the report will give: how many of the 20 have at least two valid paired F/S observations; among those with sufficient data, how many vary in F, how many vary in S, and how many vary in both; and the numerator out of 20 who simultaneously have sufficient observations and variation in F and S. Variation and the presence of high-F/low-Q responses are not pass/fail gates.

For each window with S≥2, participants will be asked for a one-sentence example and an explanation of what they considered different. Two raters, without access to numerical responses, classify each example as consistent, not consistent, or ambiguous; disagreements are resolved by consensus. Open responses do not redefine S or authorize discarding numerical ratings.

Section 6 — Criteria and decisions

Operational criteria C1, C2, and C3 retain their thresholds and functions:

- **C1 — comprehension:** at least 5 of the 6 final evaluators explain simultaneity and qualitative distinction according to the rubric and individually meet the vignette criterion (≥4/5, including one co-presence and one non-co-presence vignette).
- **C3 — recurring critical confusion:** the same confusion appears in at least 2 of the 6 final evaluators. For S, it counts as a critical confusion to treat several topics, sequence/alternation, or difference in modality, by itself, as co-presence. For F/Q, it counts as confusing future-oriented content with chain-like structure.
- **C2 — completion:** at least **240 of 300** windows have F, Q, and S completed for the same window. “Right now” probes are outside the denominator.

The pilot proceeds only if C1 is met and C3 does not occur in the final assessment. If C1 is not met or C3 occurs, there is one revision and reassessment with up to six people, within the cap of 18. If the criterion continues to fail, the current operationalization is stopped. If the pilot does not meet C2, only a documentary review of procedures follows; no second data collection is planned. No individual completion gate is created. Acceptance of the budget, release of funds, applicable ethics approval, and written confirmation of the ethics route are conditions for activities with participants.

Section 7 — Indicative timeline

The 11 months are a planning structure, not a promise about duration. Month 1: confirm the ethics route and financial conditions; month 2: applicable ethics review and finalization of materials; months 3–5: interviews, with a possible revision; month 6: freeze items and prepare administration; month 7: pilot, if authorized; months 8–9: descriptive analyses; months 10–11: feasibility report and recommendation to advance to separate planning, revise, or stop. Delays shift milestones; approval, eligibility, funding, recruitment, data collection, or completion are not guaranteed.

Section 8 — Proposed budget

The requested cap is **US$5,000**. These amounts do not indicate that funding has been received or that any call has approved it.

Category	Amount (US$)
Pilot compensation	360
Cognitive interviews	180
“Right now” probes	36
Flexible researcher support (11 months)	3,924
Miscellaneous, including transcription if needed	500
Total	**5,000**

If a fiscal sponsor is used, a fee of 5–10% of US$5,000 (US$250–500) would come out of the total, reducing flexible researcher support from US$3,924 to between US$3,424 and US$3,674. No fee has been agreed.

Section 9 — Ethics, privacy, and sharing

The route will be confirmed in writing; the cited rule provides that CONEP designates a CEP when there is no institutional affiliation, which remains to be confirmed. The reference is CNS Resolution No. 466 of 2012, item VII.2.2.

Interviews and data collection will begin only after budget acceptance, release of funds, applicable ethics approval, and written confirmation of the relevant route. The proposed age range is 18–65 years, subject to approval and final confirmation. Consent is required before participation; participants may refuse or withdraw at any time without penalty.

Responses labeled with a code, with the linkage key stored separately, are **coded/pseudonymized**, not anonymous. Any sharing of individual, raw, or identifiable data depends on consent, ethics approval, and a privacy review. Materials and results will be shared only to the extent permitted and safe. The future open-science plan (not current publication) includes the commitment: “Commitment to publish the protocol, items, vignettes, rubric, and feasibility report, even if the criteria are not met.” Individual, raw, or identifiable data are outside this commitment; any disclosure of such data depends on consent, ethics approval, and a privacy review.